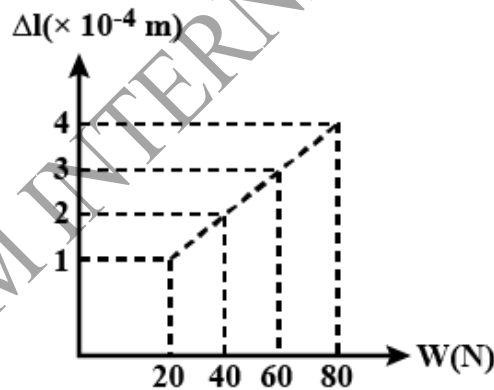


FREEDOM INTERNATIONAL SCHOOL
WORKSHEET
CLASS XI
PHYSICS
MECHANICAL PROPERTIES OF SOLIDS

1. Stress and pressure are both forces per unit area. Then in what respect does stress differ from pressure?
2. A thick wire is suspended from a rigid support, but no load is attached to its free end. Is this wire under stress?
3. Among solids, liquids and gases which possess the greatest bulk modulus?
4. Which is more elastic – rubber or steel? Explain.
5. Two wires of same length and material but of different radii are suspended from a rigid support. Both carry the same load. Will the stress, strain and extension in them be same or different?
6. A steel wire of length 4.7 m and cross-section $3.0 \times 10^{-5} \text{ m}^2$ stretches by the same amount as a copper wire of length 3.5 m and cross-section $4.0 \times 10^{-5} \text{ m}^2$ under a given load. What is the ratio of the Young's modulus of steel to that of copper?
7. A wire is cut to half its original length.
 - (a) How would it affect the elongation under a given load?
 - (b) How does it affect the maximum load it can support without exceeding the elastic limit?
8. The graph shows the extension (Δl) of a wire of length 1 m suspended from the top of a roof at one end with a load W connected to the other end. If the cross-sectional area of the wire is 10^{-6} m^2 , calculate the Young's modulus of the material of the wire.



9. If S is the stress and Y is Young's modulus of the material of a wire, what is the energy stored in the wire per unit volume in terms of S and Y ?
10. A steel wire of length 2 m is stretched through 2 mm. The cross-sectional area of the wire is 4 mm^2 . Calculate the elastic potential energy stored in the wire in the stretched condition.
Young's modulus of steel is $2 \times 10^{11} \text{ Nm}^{-2}$.